

Benjamin Curis-Friedman

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EDUCATION

McGill University

Bachelor of Software Engineering COOP 3.96 GPA

Montreal, QC

Aug. 2024 – Dec. 2027

Awards (6000\$): Alma Mater Entrance Scholarship, Class of Engineering '83 Scholarship

Stanford Online

Machine Learning Specialization by Andrew Ng 100% Grade

May 2025 – August 2025

Covering regression, deep-learning, random forests, and ML engineering practices.

EXPERIENCE

Software Developer Intern

Jan. 2026 – Present

Rogue Research Inc

Montreal, QC

- Refactored a brain surgery robot networking protocol to make it resilient to corrupt data packets.
- Developed unit tests for existing classes and methods to improve code coverage in Objective-C.
- Implemented an image comparison algorithm using vectorized math for fast processing in CI UITests.

Artificial Intelligence Intern

May 2025 – Dec. 2025

iSMART AI Lab

Montreal, QC

- Developed a video synchronization & compression library using FFMpeg and OpenCV to prepare for training.
- Coded a data pre-processing pipeline to clean raw human vital signals for machine learning models.
- Built and programmed a task labeling device using Arduino and C++ to split recording segments appropriately.
- Created a high quality multi-modal dataset of over 50 hours of recordings from dozens of study participants.

PROJECTS

AI Library From Scratch 🧠 | *Python, NumPy, TensorFlow, Keras, PyGame*

June 2025 – July 2025

- Created a NN library complete with mini-batch gradient descent and activation/cost functions.
- Trained a sequential neural network achieving ~96% test accuracy on MNIST dataset.
- Implemented customizable network and layer shapes for enhanced flexibility and scalability.
- Visualized inference with an interactive real-time digit recognition game using PyGame.

Differential Equation Simulator | *Javascript, HTML*

September 2024

- Created a web based simulation of harmonic motion.
- Programmed a visualizer and phase diagram to observe the motion of a pendulum.
- Implemented a user friendly way to set the initial conditions by clicking on the phase diagram.

ArduCar 🚗 | *Arduino, C++*

February 2024

- Built a bluetooth remote controlled car using arduino
- Designed the vehicle's body using Fusion360 to prepare for manufacturing
- Developed a motion activated controller for intuitive control using tilt sensors

WarHacks Hackathon | *Arduino, C++*

January 2024

- Engineered a fully autonomous vehicle using Arduino.
- Developed a program to follow a designated path in C++.

Web Chat Application 🗯️ | *Firebase, Javascript, Vue.js, HTML/CSS*

Oct. 2022 – Sep. 2023

- Developed a full-stack web messaging platform with authentication to message friends.
- Programmed search and adding friends features with Cloud Functions.
- Implemented Firestore security rules to ensure secure communications.
- Hosted live website on a custom domain to make it easily accessible.

SKILLS

Languages: Objective-C, C, Python, Java, Bash script, C++, Typescript, JavaScript, HTML/CSS, R

Frameworks: Next.js, React, Node.js, JUnit, Tailwind, Cocoa

Developer Tools: Git, Docker, CDash, VS Code, XCode, NeoVim, Linux, Subversion, Google Firebase

Libraries: NumPy, Matplotlib, OpenCV, pandas, SciPy, Keras, TensorFlow

Spoken Languages: English, French